

- Data transfer utilizes **JSON** format for sending analysis results.

System Features

- I. **Description:** The system must be able to read the contents of uploaded files and pasted text.
- II. **Functional Requirement 1.1:** The system shall accept .pdf and .docx file formats.
- III. **Functional Requirement 1.2:** The system shall extract text while ignoring formatting (bold, italics) and images.
- IV. **Functional Requirement 1.3:** The system shall normalize text (convert to lowercase, remove stop-words like "the", "and").

System Feature 1

4.1.1 Description and Priority

This feature handles the ingestion of user data. It is responsible for accepting the resume file (PDF/DOCX) and the Job Description (text) from the user, validating the inputs, and extracting the raw text for processing.

- **Priority: High** (The system cannot function without input).
- **Benefit: 9** (Essential for core utility).
- **Risk: 4** (Parsing errors are common with complex PDF layouts).

4.1.2 Stimulus/Response Sequences

- **User Action:** User drags and drops a file into the upload zone.
- **System Response:** System verifies file extension and size. If valid, shows a "File Uploaded" success icon. If invalid, shows an error message.
- **User Action:** User pastes text into the "Job Description" text area.
- **System Response:** System enables the "Analyze" button once both inputs are detected.

4.1.3 Functional Requirements

- **REQ-1:** The system shall allow users to register and log in using valid credentials.
- **REQ-2:** The system shall allow users to upload resume files in supported formats such as pdf
- **REQ-3:** The system shall extract text from the uploaded resume using a **resume parser**.
- **REQ-4:** The system shall identify and extract **technical and soft skills** from the resume text.
- **REQ-5:** The system shall allow users to select a **job role** or **enter job requirements** for anylse
- **REQ-6:** The system shall compare the extracted resume skills with the **required skills for the selected job role**.
- **REQ-7:** The system shall identify **missing or weak skills** based on the comparison results.
- **REQ-8:** The system shall generate **skill improvement suggestions**, such as recommended skills or courses.
- **REQ-9:** The system shall display the **skill match percentage and skill gap analysis report** to the user.
- **REQ-10:** The system shall allow the **admin to manage job roles, skill databases, and user information**.

Other Nonfunctional Requirements

Performance Requirements

Latency: The analysis process (upload to result) must complete within **5 seconds** for a standard 2-page resume.

Capacity: The system should support at least **50 concurrent users** without crashing. The system shall ensure **secure handling of uploaded resumes and user data**.

Security: The system shall ensure **secure handling of uploaded resumes and user data**.

Accuracy: The system should provide **reliable skill extraction and gap analysis**.

Usability: The system interface should be **simple and easy for job seekers to use**.

Reliability: The system should **operate continuously without failures** during normal use.

Scalability: The system should **allow addition of new job roles and skills easily**.

Safety Requirements

Error Handling: If a file is corrupted, the system must display a user-friendly error message ("File Unreadable") rather than crashing the server.

Security Requirements

- **Data Retention:** Uploaded files must be **deleted immediately** from the server after the analysis session is closed.
- **Input Validation:** All text inputs must be sanitized to prevent Cross-Site Scripting (XSS) or Injection attacks.

Software Quality Attributes

Business Rules

- **(Guest Access):** Users do not need an account to perform a resume scan.
- **(Data Privacy):** The system must not permanently store uploaded resumes; files are deleted immediately after processing.
- **(Usage Limit):** To prevent abuse, users are limited to 10 scans per hour.
- **(File Integrity):** Only text-based PDFs and DOCX files are accepted (no scanned images).
- **(Disclaimer):** The "Match Score" is an algorithmic estimate and does not guarantee employment.
- **(Admin Access):** Only Administrators can update the master skill database.

Other Requirements

Appendix A: Glossary

- **JD:** Job Description (Company requirements).
- **Parsing:** Extracting text from a file.
- **Keyword:** A specific skill (e.g., "Java") used for matching.